Chapter 9. Gaussian Linear SEMs and Directed Relations

Graphical models, Lecture 2 (DAGs, chain/fork/collider, d-separation, faithfulness, Markov equivalence, CPDAG)

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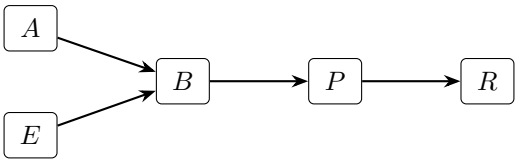
1 From Undirected to Directed Graphs

Lecture 1: conditional-dependence graph	Lecture 2: directed graph
$X_j - X_k \iff X_j \not\perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ finds relations that remain after accounting for the others edges but no directions	$X_k \to X_j$ expresses each arrow by a generating equation per variable reads the conditional independences the graph creates; checks which directions data cannot distinguish

Question: **how far can directions be inferred from Gaussian observational data?**

2 A Gaussian Linear SEM Is a Bundle of Regressions on a DAG

2.1 The health example as equations



$$A = \epsilon_A, \quad E = \epsilon_E, \quad B = \beta_{AB}A + \beta_{EB}E + \epsilon_B, \quad P = \beta_{BP}B + \epsilon_P, \quad R = \beta_{PR}P + \epsilon_R,$$
$$\epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma_j^2).$$

Definition: DAG and Gaussian linear SEM

A directed acyclic graph (DAG) G has directed edges and no directed cycle. With $\text{pa}(j)$ the parents of node j,

$$X_j = \sum_{k \in \text{pa}(j)} \beta_{kj} X_k + \epsilon_j, \qquad \epsilon_1, \ldots, \epsilon_p \text{ independent}, \ \epsilon_j \sim N(0, \sigma_j^2).$$

Variable	Parents	Local regression
A, E	none	$A = \epsilon_A$, $E = \epsilon_E$
B	A, E	$B = \beta_{AB}A + \beta_{EB}E + \epsilon_B$
P	B	$P = \beta_{BP}B + \epsilon_P$
R	P	$R = \beta_{PR}P + \epsilon_R$

E is not in the equation of P but is related to P through $E \to B \to P$.

2.2 Matrix form and the implied covariance

Let $B_{jk} = \beta_{kj}$ and $D = \text{diag}(\sigma_1^2, \ldots, \sigma_p^2)$. Then $X = BX + \epsilon$; acyclicity makes B permutable to strictly lower triangular, so

$$X = (I - B)^{-1}\epsilon, \qquad \Sigma = (I - B)^{-1} D (I - B)^{-\top}, \qquad \Omega = \Sigma^{-1} = (I - B)^{\top} D^{-1} (I - B).$$

Markov factorization:

$$p(x_1, \ldots, x_p) = \prod_{j=1}^{p} p(x_j \mid x_{\text{pa}(j)}).$$

Theorem: the GGM of an SEM is its moral graph

For $j \neq k$,

$$\Omega_{jk} = -\frac{B_{jk}}{\sigma_j^2} - \frac{B_{kj}}{\sigma_k^2} + \sum_{l \neq j,k} \frac{B_{lj} B_{lk}}{\sigma_l^2}.$$

Hence $\Omega_{jk} \neq 0$ only if j, k are adjacent in G or have a common child; generically (no cancellation) exactly then. The GGM of Ch. 8 is the **moral graph**: the skeleton plus edges "marrying" the parents of each child.

Proof [Supplement]. $\Omega_{jk} = \sum_l (I-B)_{lj}(I-B)_{lk}/\sigma_l^2$: $l = j$ gives $-B_{jk}/\sigma_j^2$, $l = k$ gives $-B_{kj}/\sigma_k^2$, $l \neq j, k$ gives $B_{lj}B_{lk}/\sigma_l^2$, nonzero only for a common child l. ■

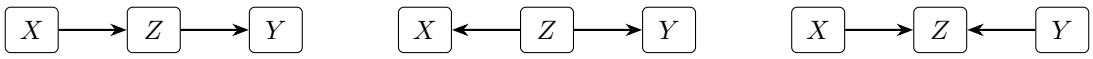
In the health example $A \perp\!\!\!\perp E$, yet the GGM has $A - E$ (common child B).

3 Chains, Forks and Colliders

Motivation. Chains and forks transmit association unless the middle node is conditioned on; colliders do the opposite.

Structure	Graph	Marginally	Conditioning on Z
chain	$X \to Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
fork	$X \leftarrow Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
collider	$X \to Z \leftarrow Y$	$X \perp\!\!\!\perp Y$	X and Y related

(Under Markov and faithfulness, §4.) Conditioning **closes** chains and forks and **opens** colliders.



chain: open given Z: blocked | fork: open given Z: blocked | collider: blocked given Z: open

Figure 1: The three elementary paths and the effect of conditioning on Z. Only the collider reverses, so v-structures are detectable; conditioning on a descendant of a collider also opens it.

Theorem: the three patterns in a Gaussian SEM

Let all errors be independent with variances 1 and coefficients $a, b \neq 0$.

1. Chain $Z = aX + \epsilon_Z$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.
2. Fork $X = aZ + \epsilon_X$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.
3. Collider $Z = aX + bY + \epsilon_Z$ with $X \perp\!\!\!\perp Y$: $\mathrm{Cov}(X, Y) = 0$ but $\rho_{XY \cdot Z} = -\dfrac{ab}{\sqrt{(1+b^2)(1+a^2)}} \neq 0$.

Proof [Supplement]. (1) $Y = abX + b\epsilon_Z + \epsilon_Y$, so $\mathrm{Cov}(X, Y) = ab$; the residual of Y on Z is $\epsilon_Y \perp\!\!\!\perp (X, Z)$.

(2) $\mathrm{Cov}(X, Y) = ab\mathrm{Var}Z$; residuals on Z are ϵ_X, ϵ_Y, independent.

(3) $\mathrm{Cov}(X, Y) = 0$. With $\mathrm{Var}Z = a^2 + b^2 + 1$, $\mathrm{Cov}(X, Z) = a$, $\mathrm{Cov}(Y, Z) = b$: residual covariance $-ab/(a^2 + b^2 + 1)$, residual variances $(1 + b^2)/(a^2 + b^2 + 1)$ and $(1 + a^2)/(a^2 + b^2 + 1)$; divide. ■

Collider "explaining away": high BMI and low age suggest low exercise. **This change of conditional independence distinguishes a collider.**

4 d-Separation, the Markov Condition and Faithfulness

Definition: d-separation

A path between X and Y is **blocked** by a set S if it contains (i) a non-collider (chain or fork node) that is in S, or (ii) a collider that is not in S and has no descendant in S. X and Y are d-separated by S, written $X \perp_G Y \mid S$, if every path between them is blocked.

Definition: causal Markov condition and faithfulness

Markov: $X \perp_G Y \mid S \Rightarrow X \perp\!\!\!\perp Y \mid S$ in the distribution (equivalent to the Markov factorization). **Faithfulness**: the converse, $X \perp\!\!\!\perp Y \mid S \Rightarrow X \perp_G Y \mid S$: every conditional independence in the distribution is implied by the graph.

Under both, CIs of the data are *exactly* the d-separations of G (used in Ch. 10).

Example: how faithfulness fails — cancellation

$X \sim N(0,1)$, $Z = aX + \epsilon_Z$, $Y = cX + bZ + \epsilon_Y$: $\mathrm{Cov}(X,Y) = c + ab$. If $c = -ab$, $X \perp\!\!\!\perp Y$ despite the edge $X \to Y$. Such values have Lebesgue measure zero, but near-cancellation makes finite-sample tests unreliable.

Example: d-separation in the health example

$A \perp E$ (the only path $A \to B \leftarrow E$ has an unconditioned collider), but $A \not\perp E \mid P$ (P is a descendant of the collider B). $E \perp R \mid B$ and $E \perp R \mid P$ (the chain is blocked). $A \perp P \mid B$.

5 Markov Equivalence

5.1 Directions with the same conditional independences cannot be distinguished

$$X_1 \to X_2 \to X_3, \qquad X_1 \leftarrow X_2 \to X_3, \qquad X_1 \leftarrow X_2 \leftarrow X_3$$

all imply only $X_1 \perp\!\!\!\perp X_3 \mid X_2$. DAGs with the same CIs form a **Markov equivalence class**.

Theorem: the three DAGs define the same Gaussian model

A Gaussian distribution on (X_1, X_2, X_3) factorizes according to one of the three DAGs iff $X_1 \perp\!\!\!\perp X_3 \mid X_2$; so all three describe exactly the same set of covariance matrices.

Proof [Supplement]. If $X_1 \perp\!\!\!\perp X_3 \mid X_2$, $p = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_2)$ (chain); Bayes $p(x_1)p(x_2 \mid x_1) = p(x_2)p(x_1 \mid x_2)$ gives the fork, and $p(x_2)p(x_3 \mid x_2) = p(x_2 \mid x_3)p(x_3)$ the reversed chain. Conversely each factorization implies the CI. Gaussian conditionals are linear regressions, so each factorization is a Gaussian linear SEM. ■

The collider $X_1 \to X_2 \leftarrow X_3$ ($X_1 \perp\!\!\!\perp X_3$, $X_1 \not\perp\!\!\!\perp X_3 \mid X_2$) *is* distinguishable.

5.2 Skeleton and unshielded colliders

Definition: skeleton and v-structure

The **skeleton** is the adjacency structure left after deleting arrowheads: $X \to Z \to Y \mapsto X - Z - Y$. An **unshielded collider** (v-structure) is $X \to Z \leftarrow Y$ with X and Y non-adjacent.

Theorem: Verma–Pearl characterization

Two DAGs are Markov equivalent iff they have the same skeleton and the same unshielded colliders.

Idea [Supplement]. X, Y adjacent iff no set d-separates them (non-adjacent: the parents of the later one in a topological order separate). Given the skeleton, an unshielded $X - Z - Y$ is a collider iff Z is in *no* separating set of X, Y. Conversely, equal skeletons and v-structures give the same blocking sets on every path. ■

5.3 CPDAG

Definition: CPDAG

The completed partially directed acyclic graph of an equivalence class has a directed edge $X \to Y$ if every DAG in the class contains $X \to Y$, and an undirected edge $X - Y$ if the direction varies within the class.

CPDAG $X - Z - Y$ represents the chain, fork and reversed chain; the collider's CPDAG keeps both arrows. Health example: CPDAG $A \to B \leftarrow E$, $B \to P \to R$ ($B \to P$ forced, else a new v-structure $A \to B \leftarrow P$; likewise $P \to R$). **An undirected edge means an undistinguished direction, not "no relation".**

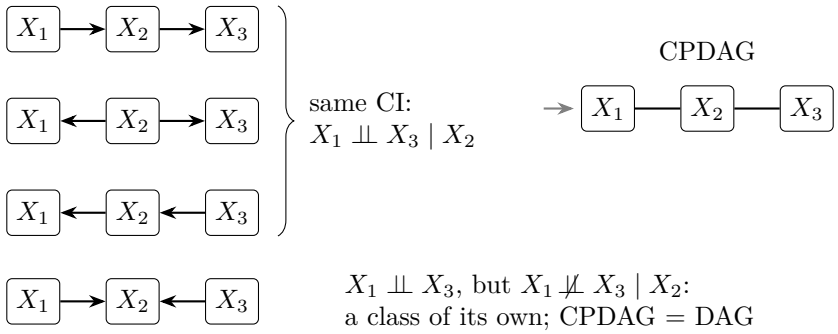


Figure 2: Markov equivalence: chain, fork and reversed chain share skeleton and CIs (undirected CPDAG); the collider is identified.

Summary

- Gaussian linear SEM: $\Sigma = (I - B)^{-1} D (I - B)^{-\top}$; its GGM is the moral graph.
- Conditioning closes chains and forks, opens colliders (d-separation).
- Markov + faithfulness: CIs = d-separations.
- Markov-equivalent DAGs share skeleton and v-structures; data identify only the CPDAG.

6 Selected exercises

Motivation. Structural equations as parent regressions, conditioning on a collider's descendant, and the equivalence class of a path.

Exercise 9.1 (Structural errors and non-descendants)

In a Gaussian linear SEM on a DAG G, let $\mathrm{nd}(j)$ be the set of non-descendants of j other than j.
(a) Show that ϵ_j is independent of $X_{\mathrm{nd}(j)}$.
(b) Deduce that the population least-squares regression of X_j on $X_{\mathrm{pa}(j)}$ has coefficients β_{kj} and residual variance σ_j^2, and that $X_j \perp\!\!\!\perp X_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)} \mid X_{\mathrm{pa}(j)}$ (local Markov property).

Solution [Supplement].

(a)

Step 1. In topological order B is strictly lower triangular, $B^p = 0$, $(I-B)^{-1} = \sum_{m=0}^{p-1} B^m$; $(B^m)_{kl}$ sums edge-coefficient products over directed paths $l \to \cdots \to k$ of length m, so $\{(I-B)^{-1}\}_{kl} \neq 0$ only if $l = k$ or l is an ancestor of k.

Step 2. For $k \in \mathrm{nd}(j)$, $X_k = \sum_l \{(I-B)^{-1}\}_{kl}\epsilon_l$ involves only $(\epsilon_l)_{l\neq j}$, which are independent of ϵ_j; so $\epsilon_j \perp\!\!\!\perp X_{\mathrm{nd}(j)}$.

(b)

Step 3. $\mathrm{pa}(j) \subseteq \mathrm{nd}(j)$ (else a directed cycle). With $X_P = X_{\mathrm{pa}(j)}$, $\beta = (\beta_{kj})_{k\in\mathrm{pa}(j)}$: $\mathbb{E}(X_P\epsilon_j) = 0$, so β solves $\mathbb{E}\{X_P(X_j - X_P^\top\beta)\} = 0$, uniquely since $\mathrm{Var}(X_P)$ is a principal submatrix of $\Sigma \succ 0$. Residual ϵ_j, variance σ_j^2.

Step 4. With $X_N = X_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)}$: given $(X_P, X_N) = (x_P, x_N)$, $X_j = x_P^\top\beta + \epsilon_j$ with $\epsilon_j \perp\!\!\!\perp (X_P, X_N)$ by (a), so the law $N(x_P^\top\beta, \sigma_j^2)$ is free of x_N: $X_j \perp\!\!\!\perp X_N \mid X_P$. ■

Exercise 9.2 (Conditioning on a descendant of a collider)

Let $X, Y, \epsilon_Z, \epsilon_W$ be independent $N(0,1)$, $Z = aX + bY + \epsilon_Z$ and $W = cZ + \epsilon_W$ with $a, b, c \neq 0$. Show that

$$\rho_{XY\cdot W} = -\frac{c^2ab}{\sqrt{\{c^2(1+b^2)+1\}\{c^2(1+a^2)+1\}}} \neq 0,$$

and that $|\rho_{XY\cdot W}|$ is strictly smaller than the collider value $|\rho_{XY\cdot Z}| = |ab|/\sqrt{(1+a^2)(1+b^2)}$.

Solution [Supplement].

Step 1. $\mathrm{Cov}(X,W) = ca$, $\mathrm{Cov}(Y,W) = cb$, $v := \mathrm{Var}W = c^2(a^2+b^2+1)+1$.

Step 2. Residuals on W: $X - \frac{ca}{v}W$, $Y - \frac{cb}{v}W$, with covariance $-\frac{c^2ab}{v}$ and variances $1 - \frac{c^2a^2}{v} = \frac{c^2(1+b^2)+1}{v}$, $\frac{c^2(1+a^2)+1}{v}$. Divide; nonzero as $a, b, c \neq 0$.

Step 3. Dividing by c^2,

$$|\rho_{XY\cdot W}| = \frac{|ab|}{\sqrt{(1+b^2+c^{-2})(1+a^2+c^{-2})}} < \frac{|ab|}{\sqrt{(1+a^2)(1+b^2)}}.$$

W opens $X \to Z \leftarrow Y$, less than Z; the collider value is reached as $c^2 \to \infty$. ■

Exercise 9.3 (The equivalence class of a path)

Consider DAGs whose skeleton is the path $X_1 - X_2 - \cdots - X_p$, $p \geq 2$. Show that exactly p of them have no v-structure, namely those with a single source X_s,

$$X_1 \leftarrow \cdots \leftarrow X_{s-1} \leftarrow X_s \rightarrow X_{s+1} \rightarrow \cdots \rightarrow X_p, \qquad s = 1, \ldots, p,$$

that these form one Markov equivalence class, and that its CPDAG is the undirected path.

Solution [Supplement].

Step 1. Every orientation of a tree is acyclic. Let $e_i = \mathsf{R}$ if $X_i \rightarrow X_{i+1}$, $e_i = \mathsf{L}$ otherwise.

Step 2. Only consecutive edges share a node; (X_{i-1}, X_i, X_{i+1}) is unshielded and a collider iff $e_{i-1}e_i = \mathsf{RL}$. So no v-structure iff $e_1 \cdots e_{p-1}$ has no factor RL.

Step 3. If $e_i = \mathsf{R}$, $e_j = \mathsf{L}$ with $i < j$, let i' be the largest index $< j$ with $e_{i'} = \mathsf{R}$; then $e_{i'+1} = \mathsf{L}$, an RL. Hence words without RL are exactly $\mathsf{L}^{s-1}\mathsf{R}^{p-s}$, $s = 1, \ldots, p$: the single-source DAGs.

Step 4. By Verma–Pearl these p DAGs are equivalent, and any equivalent DAG has the same skeleton and no v-structure, so is one of them. e_i is R for $s \leq i$ and L for $s > i$, so every edge varies: the CPDAG is undirected. ■